2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall) | 979-8-3315-0320-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/VTC2025-Fall65116.2025.11309847

A Pseudo-Increment Prediction Method for GNSS/INS Integrated Navigation Based on Transformer–LSTM During GNSS Outages

Jin Wang[1], Siyuan Xu[1], Yan Xing[2], Jin Lu[1], Huimin Hu [1]
[1]School of Communication and Information Engineering, Xi'an University of Posts and Telecommunications, Xi'an 710121, China
[2] National Time Service Center, Chinese Academy of Sciences, Xi'an 710600, China
Emails:wangjin@xupt.edu.cn, siyuanxu@stu.xupt.edu.cn, xingyan@ntsc.ac.cn, lujin@xupt.edu.cn, huiminhu@xupt.edu.cn

***Abstract*—With the rapid development of intelligent unmanned systems and autonomous navigation technologies, particularly driven by the strategic demands of the low-altitude economy, navigation systems are facing increasingly higher requirements for accuracy and robustness. During global navigation satellite system(GNSS) outages, GNSS/INS integrated navigation degrades into standalone inertial navigation system(INS), which leads to rapid accumulation of positioning errors. This paper proposes the GNSS pseudo-increment prediction method based on the Transformer–long short-term memory (LSTM) network. It utilizes inertial measurement unit measurements and INS output navigation results as inputs, where the Transformer is employed to extract global temporal features and the LSTM is used to model temporal dependencies. The predicted GNSS pseudo-increments are then fused with INS outputs via the Kalman filter(KF) for state updates. Experiments based on the publicly available vehicle dataset demonstrate that during 50s GNSS outage, the proposed method achieves over 90% reduction in horizontal errors compared to pure INS, and over 40% compared with LSTM and Transformer models, respectively. These results confirm that the method effectively mitigates INS drift and significantly enhances the accuracy and robustness of navigation and positioning.**

***Index Terms*—GNSS/INS integrated navigation, Transformer-LSTM, pseudo-increment prediction, GNSS outage, Kalman filtering**

I. Introduction

GNSS/INS integrated navigation systems have been widely applied in unmanned platforms and autonomous vehicles, particularly in low-altitude economy (LAE) scenarios, due to their ability to combine the global coverage of the global navigation satellite system(GNSS) with the high-frequency, disturbance-resistant outputs of inertial navigation system(INS). This fusion improves navigation accuracy and robustness [1], typically achieved through Kalman filtering(KF). However, in environments like urban canyons or tunnels, GNSS signals are often blocked, leading to delayed or missing KF updates. Consequently, the system reverts to standalone INS, and positioning accuracy rapidly deteriorates due to accumulated inertial the measurement unit(IMU) errors [2].

To address the degradation of navigation stability during GNSS outages, recent studies have explored the use of neural network models to establish nonlinear mappings between IMU outputs and navigation states [3]. For example, Sharaf et al. [4] employed the radial basis function neural network to compensate for the loss of GNSS signals by taking INS output position outputs as inputs and training the network to predict the position differences between INS and GNSS. El-Sheimy et al. [5] proposed a multilayer perceptron (MLP) based position update framework to predict GNSS pseudo positions and integrate them with KF to enhance positioning performance. However, MLP networks are time consuming to train, and their internal structure and parameter tuning are often difficult to determine. To address this, Xu et al. [6] employed a backpropagation neural network to predict GNSS pseudo-increments, constrain GNSS/INS integration, and enable real-time outlier correction, thereby enhancing positioning stability during GNSS outages.However, these three models belong to static neural networks, which are incapable of fully leveraging historical data for sequential positioning estimation. To address this limitation, Fang et al. [7] introduced a long short-term memory (LSTM) network for GNSS pseudo increment prediction, which effectively suppresses error accumulation in standalone INS navigation. LSTM can capture long-range dependencies in the temporal domain. Building upon this work, Zhi et al. [8] introduced a convolutional neural network (CNN) to construct a CNN–LSTM hybrid architecture, where CNN was utilized to extract temporal features from IMU data and enhance the learning capacity and predictive accuracy of the LSTM. Further extending this approach, Chen et al. [9] incorporated an attention mechanism and proposed an Attention–LSTM based error compensation model. By applying dynamic attention weights, the model is able to focus on IMU features at critical time steps, thereby improving the LSTM's capacity to model navigation states with greater precision.

To address the limitations of existing methods in feature extraction and temporal modeling, this paper proposes a pseudo-increment prediction method based on Transformer–LSTM network. The model takes IMU measurements and INS derived outputs as input, with GNSS position and velocity increments as learning targets. By integrating the global attention mechanism of the Transformer and the temporal modeling capability of LSTM, the model captures both long-range dependencies and local temporal dynamics. By enhancing the network's

This work was supported in part by the Natural Science Basic Research Program of Shaanxi Province under Grant 2025JC-YBMS-771 and the National Natural Science Foundation of China under Grant 62371392.

979-8-3315-0320-8/25/$31.00 ©2025 IEEE
Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 25,2026 at 23:28:21 UTC from IEEE Xplore. Restrictions apply.

capability to capture complex temporal dependencies, this complementary architecture improves both training efficiency and prediction performance, and ensures greater continuity and robustness of integrated navigation during GNSS signal interruptions.

The remainder of this paper is structured as follows: Section II outlines the GNSS/INS integration and network modules. Section III details the proposed framework and Transformer–LSTM model. Section IV reports experimental validation, and Section V concludes the paper.

II. Mathematical model

A. GNSS/INS Loosely Coupled Model

In a loosely coupled GNSS/INS integration structure, the GNSS navigation solution is treated as the observation and combined with the velocity and position estimates derived from the INS to serve as input to the Kalman filter for system state updates. This structure offers several advantages, including simplicity, low computational cost, and ease of engineering implementation [10]. In this configuration, the system state error closely approximates the accumulated error of the INS, which continuously output navigation solutions attitude angle A_{INS}, velocity V_{INS}, position P_{INS} in real time. GNSS provides position and velocity information, which, when differenced with INS output, is used to construct the observation vector Z_k for the KF. The Kalman filter module updates the system state through prediction process, yielding estimates of attitude error δA, velocity error δV, and position error δP. These error estimates are fed back to correct the INS solution in a closed loop manner, resulting in the final integrated navigation output [11].

The standard KF is employed to recursively estimate the system state vector. The state vector consists of 15 components, which are defined as follows:

$$X = \begin{bmatrix} \phi_e & \phi_n & \phi_u & \delta V_e & \delta V_n & \delta V_u & \delta L & \delta\lambda & \delta h \\ & & & \nabla_x & \nabla_y & \nabla_z & \varepsilon_x & \varepsilon_y & \varepsilon_z \end{bmatrix}^T \quad (1)$$

Show above, ϕ_e, ϕ_n and ϕ_u denotes the attitude error in the east, north and up directions, velocity error along these axes are represented by $\delta V_e, \delta V_n$ and δV_u, $\delta L, \delta\lambda$ and δh corresponds to the position error in terms of latitude, longitude, and altitude, $[\nabla_x\, \nabla_y\, \nabla_z]$ and $[\varepsilon_x\, \varepsilon_y\, \varepsilon_z]$ represent the accelerometer biases and gyroscope biases in three axes.

The observation vector Z_k is:

$$Z_k = \begin{bmatrix} P_{GNSS} - P_{INS} \\ V_{GNSS} - V_{INS} \end{bmatrix} \quad (2)$$

Based on the above description, the state equation and observation equation of the Kalman Filter are constructed as follows:

$$\begin{cases} X_k = F_{k,k-1} X_{k-1} + G_k W_k \\ Z_k = H_k X_k + V_k \end{cases} \quad (3)$$

where $F_{k,k-1}$ represents the state transition matrix, G_k is the system noise distribution matrix, and H_k represents the observation matrix. W_k and V_k denote the process noise and observation noise, respectively.

The KF process consists of the prediction and update step [12], described in detail below:

Prediction stage:

$$\hat{X}_{k,k-1} = F_{k,k-1} \hat{X}_{k-1} \quad (4)$$

$$P_{k,k-1} = F_{k,k-1} P_{k-1} F_{k,k-1}^T + G_{k,k-1} Q_{k-1} G_{k,k-1}^T \quad (5)$$

Update stage:

$$K_k = P_{k,k-1} H_k^T (H_k P_{k,k-1} H_k^T + R_k)^{-1} \quad (6)$$

$$\hat{X}_k = \hat{X}_{k,k-1} + K_k (Z_k - H_k \hat{X}_{k,k-1}) \quad (7)$$

$$\hat{P}_k = (I - K_k H_k) P_{k,k-1} \quad (8)$$

where $\hat{X}_k$ denotes the updated state estimate, and $\hat{P}_k$ is the updated state covariance matrix, K_k represents the Kalman gain, Q_k corresponds to the process noise covariance matrix, while R_k refers to the observation noise covariance matrix.

B. Transformer structure

The Transformer employs the self-attention mechanism for feature extraction by constructing the query(Q), key(K), and value(V) matrices to compute attention weights between different positions. The computation is formulated as follows:

$$Attention(Q, K, V) = Soft\max(\frac{QK^T}{\sqrt{d_k}})V \quad (9)$$

where $Soft\max(\cdot)$ denotes the normalized exponential function, and d_k is the dimensionality of the key vectors. The scaling factor $\sqrt{d_k}$ is applied to prevent gradient explosion or vanishing.

The multi-head attention mechanism captures feature dependencies across different subspaces by computing multiple self-attention heads in parallel [13]. The computation process is as follows:

$$MultiHead(Q, K, V) = Concat(head_1, ..., head_h) W^o \quad (10)$$

where each attention head is defined by:

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (11)$$

where $Concat(\cdot)$ represent the concatenation operation, W_i^Q, W_i^K, W_i^V, W^o are the learnable weight matrices for the query, key, value, and output projections, respectively.

C. LSTM structure

LSTM network addresses the gradient vanishing and explosion problems commonly encountered in traditional recurrent neural network when processing long sequences, through its gating mechanism. The internal structure includes three gates: i_t for the input gate, f_t for the forget gate, and o_t for the output gate. The core computation is formulated as follows [14]:

$$\begin{pmatrix} f_t \\ i_t \\ o_t \\ \hat{C}_t \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} + b \quad (12)$$

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 25,2026 at 23:28:21 UTC from IEEE Xplore. Restrictions apply.

$$C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t \tag{13}$$

$$h_t = o_t \odot \tanh(C_t) \cdot \tag{14}$$

where h_t denotes the hidden state, $\hat{C}_t$ represents the candidate cell state, σ and $tanh$ refer to the sigmoid and hyperbolic tangent activation functions, respectively, W and b correspond to the weight matrices and bias vectors, and $\odot$ denotes the element-wise multiplication.

III. Methodology

A. Overall System Architecture Design

Existing AI-assisted models addressing navigation accuracy degradation during GNSS outages can be broadly categorized into three types: $O_{INS} - \delta P_{INS}$ model, $O_{INS} - X_k$ model, and $O_{INS} - \Delta P_{GNSS}$ model [15].The first two types typically use INS outputs as inputs to neural networks, aiming to predict either the integrated navigation errors or the dynamic evolution of the KF state vector. Although these approaches exhibit certain compensation capabilities during GNSS interruptions, they share a common drawback: the prediction results inevitably incorporate the inherent errors of the INS and the observation noise from GNSS, which may introduce additional mixed-error interference into the overall system.

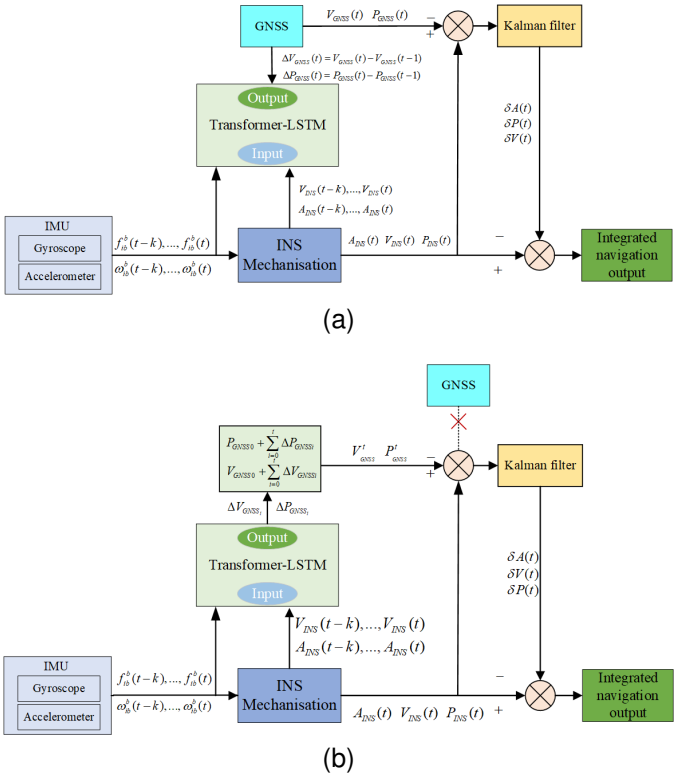


Fig. 1. Architecture of the GNSS/INS integrated navigation system based on Transformer–LSTM, (a) GNSS available phase, (b) GNSS outages phase

To overcome the aforementioned issues, this paper proposes an integrated navigation compensation method based on pseudo increment prediction, which takes GNSS output position and velocity increments as the prediction targets. The Transformer–LSTM network is adopted to avoid the fusion of multiple sources of error. In the GNSS available area as shown in Fig. 1(a), the system takes IMU measured angular velocity ω_{ib} and acceleration f_{ib}, and INS derived output velocity V_{INS} and attitude A_{INS}, as the input to the neural network. The target outputs are the GNSS position increment ΔP_{GNSS} and velocity increment ΔV_{GNSS}. Fig. 1(b) shows the prediction mode activated during GNSS outages. The trained Transformer–LSTM network takes the input features and predicts pseudo-increment information, which is used to generate pseudo GNSS observations. These observations are then fed into the Kalman filter to update the system state and continuously correct the INS derived navigation solution.

B. Transfomer-LSTM network model

This paper presents a GNSS pseudo-increment estimation model constructed upon a Transformer–LSTM architecture. As depicted in Fig. 2, the model adopts a cascaded framework composed of an input feature layer, a Transformer encoder, an LSTM layer, and a regression output module.Initially, the input undergoes dimensional transformation and positional encoding, and then passed into the Transformer encoder, where multi-head attention is utilized to capture temporal dependencies ranging from short-term to long-term scales. The extracted features are delivered to the LSTM unit for further modeling of sequential correlations. Finally, a fully connected network performs regression on the LSTM output to estimate the pseudo GNSS position and velocity increments along the east and north axes at the current epoch.

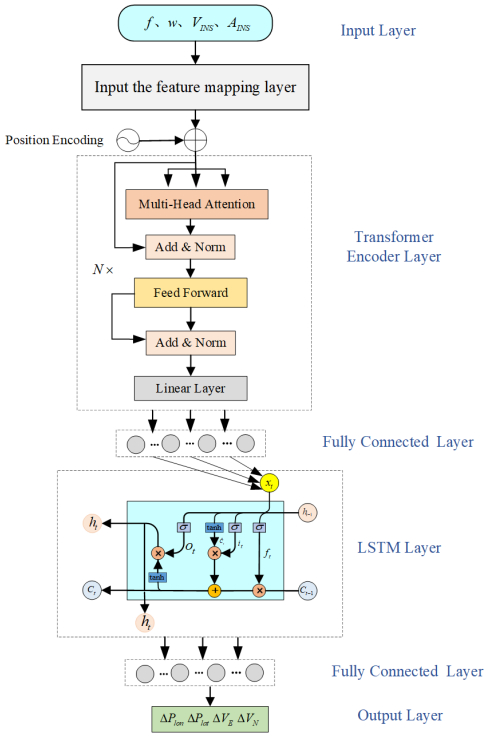


Fig. 2. Structural design of the Transformer–LSTM network

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 25,2026 at 23:28:21 UTC from IEEE Xplore. Restrictions apply.

IV. Experiments and Analysis

The experimental data applied in this study were collected by Prof. Gongmin Yan at Northwestern Polytechnical University. This dataset is publicly accessible and available for download at www.psins.org.cn. The total duration of the dataset is 3170s. The IMU employed is the STIM300, with the sampling frequency of 125 Hz, while the GNSS delivers single-point positioning updates at 1 Hz. The IMU parameter settings are shown in Table I.

TABLE I
STIM300 Performance Parameters

Parameter settings	Parameter values
Bias of gyroscope	100
Gyroscope gaussian noise	0.1
Bias of accelerometer	5000
Accelerometer gaussian noise	10ug

To evaluate the navigation compensation capability of the Transformer–LSTM model under GNSS outage conditions, the first 1000s of the dataset were selected. The initial 800s were used for model training, and the subsequent period from 800 to 1000s was used for testing. A GNSS outage was simulated from 800s to 850s, during which the vehicle trajectory included a turning segment to create a challenging navigation scenario. As depicted in Fig. 3, the blue path reflects the entire GNSS trajectory, while the red segment corresponds to the simulated outage interval. The Transformer–LSTM model

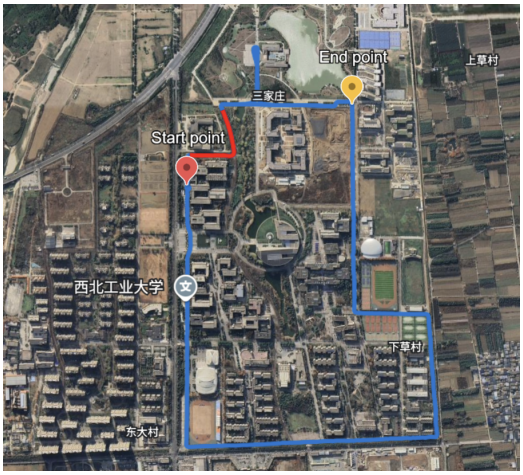


Fig. 3. Vehicle trajectory

takes as input the 12-dimensional feature vector consisting of three-axis gyroscope data, three-axis accelerometer data, velocity, and attitude vectors. The output comprises four dimensional GNSS pseudo-increments, representing position and velocity changes in the east and north directions.

Fig. 4 compares the navigation trajectories of different methods during 50s GNSS outage.The majority of the error deviations are primarily distributed within the turning phase. The black trajectory represents the reference, while the red trajectory, corresponding to the proposed Transformer–LSTM

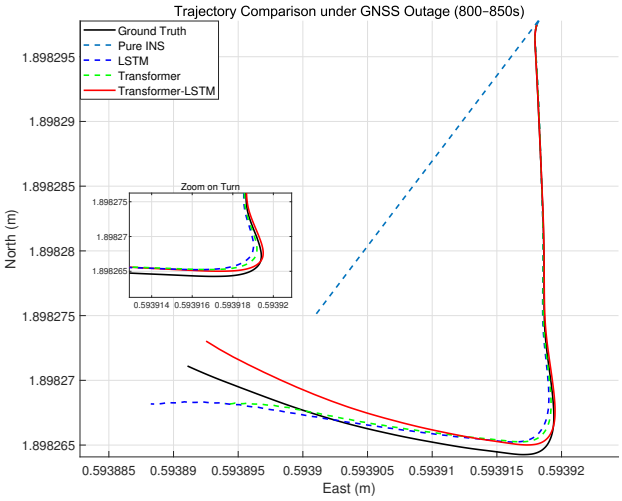


Fig. 4. Trajectories estimated by different methods during GNSS outage

method, maintains high consistency with the reference in both overall accuracy and turning behavior, outperforming the other methods. The Transformer exhibits slight drift in the later phase of the outage, whereas the LSTM shows noticeable deviation during the path change. The Pure INS trajectory deviates significantly. The inset provides a zoomed-in view of the turning segment.

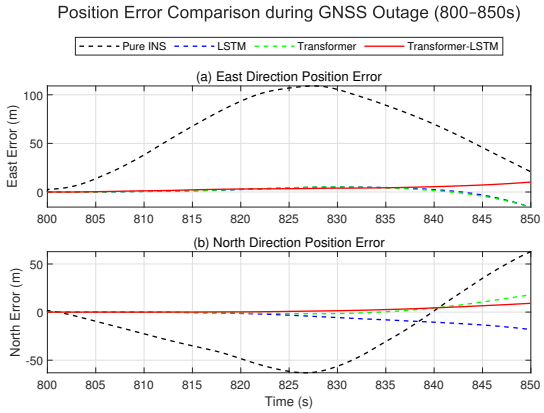


Fig. 5. Position errors under different methods

Fig. 5 and 6 illustrate the variations in position and velocity errors in the east and north directions, respectively, for each method during the GNSS outage. Horizontal error is defined as the combined error in the east and north directions, reflecting planar positioning accuracy.The results demonstrate that, without GNSS observations, the Pure INS (PINS) method suffers from rapid error accumulation and divergence. Compared with PINS, LSTM, and Transformer, the proposed Transformer–LSTM model achieves reductions in the maximum position error along the east direction by 90.67%, 34.30%, and 33.70%, respectively, and in the north direction by 85.51%, 49.34%, and 49.26%, respectively.For

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 25,2026 at 23:28:21 UTC from IEEE Xplore. Restrictions apply.

maximum velocity errors, the Transformer–LSTM achieves reductions of 94.11% and 94.20% in the east and north directions compared to PINS, 45.45% and 34.92% compared to LSTM, and 41.94% and 45.33% compared to Transformer. The statistical results of positioning and velocity errors for different methods during the GNSS outage are summarized in Tables II and III.

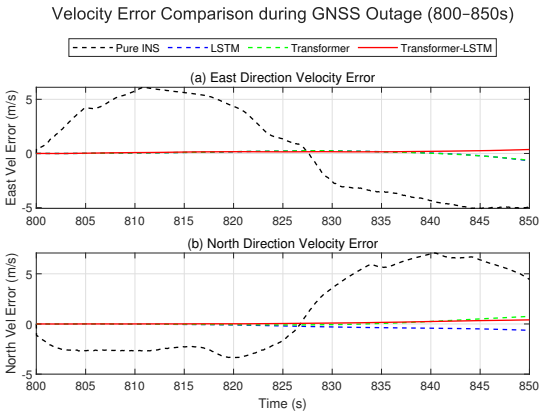


Fig. 6. Velocity errors under different methods

TABLE II
MAXIMUM ERRORS IN POSITION AND VELOCITY ACROSS METHODS

Method	Direction	Position Max (m)	Velocity Max (m/s)
PINS	East	109.11	6.11
	North	63.09	7.07
	Horizontal	125.88	8.39
LSTM	East	15.49	0.66
	North	18.05	0.63
	Horizontal	23.79	0.92
Transformer	East	15.35	0.62
	North	18.02	0.75
	Horizontal	23.68	0.97
Transformer–LSTM	East	10.18	0.36
	North	9.14	0.41
	Horizontal	13.68	0.54

TABLE III
POSITION AND VELOCITY RMSE COMPARISON ACROSS METHODS

Method	Direction	Position RMSE (m)	Velocity RMSE (m/s)
PINS	East	72.68	4.18
	North	38.82	4.23
	Horizontal	82.39	5.95
LSTM	East	4.14	0.19
	North	7.39	0.30
	Horizontal	8.47	0.35
Transformer	East	4.22	0.19
	North	5.23	0.24
	Horizontal	6.72	0.31
Transformer–LSTM	East	4.37	0.17
	North	3.31	0.17
	Horizontal	5.48	0.24

V. CONCLUSION

In conclusion, this paper proposes a GNSS pseudo increment prediction method based on the Transformer–LSTM network to improve the continuity and accuracy of GNSS/INS integrated navigation under GNSS outage conditions.. The method combines the global feature extraction capability of the Transformer with the temporal modeling strength of LSTM to predict pseudo GNSS increments, which are incorporated into a Kalman filter for navigation state updates. Experimental results on a public vehicle dataset demonstrate the proposed approach's effectiveness in enhancing positioning accuracy and robustness. The approach requires no external auxiliary information, offers good computational efficiency and generalization capability, and continuously enhances navigation continuity under GNSS outages. It is well suited for intelligent perception and autonomous navigation tasks of multi-agent systems operating in LAE environments.

REFERENCES

[1] Y. He, J. Li, and J. Liu, "Research on GNSS INS and GNSS/INS integrated navigation method for autonomous vehicles: A survey," *IEEE Access*, vol. 11, pp. 79033–79055, 2023.

[2] Y. Gao, X. Yuan, D. Yang, Y. Hu, Y. Cao, and A. Schmeink, "UAV-Assisted MEC System with Mobile Ground Terminals: DRL-Based Joint Terminal Scheduling and UAV 3D Trajectory Design," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 1–17, Jan. 2024.

[3] J. Du, X. Li, H. Liu, J. Wang, and Y. Liu, "QoE-Guaranteed Optimization in MEC-Enabled Metaverse: An Active Inference Deep Reinforcement Learning Approach," *IEEE Trans. Cogn. Commun. Netw.*, early access, Feb. 2025, doi: 10.1109/TCCN.2025.3554003.

[4] R. Sharaf, A. Noureldin, A. Osman, and N. El-Sheimy, "Online INS/GPS integration with a radial basis function neural network," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 20, no. 3, pp. 8–14, 2005.

[5] N. El-Sheimy, K. W. Chiang, and A. Noureldin, "The utilization of artificial neural networks for multisensor system integration in navigation and positioning instruments," *IEEE Trans. Instrum. Meas.*, vol. 55, no. 5, pp. 1606–1615, 2006.

[6] Y. Xu, K. Wang, C. Jiang, Z. Li, C. Yang, D. Liu, and H. Zhang, "Motion-constrained GNSS/INS integrated navigation method based on BP neural network," *Remote Sens.*, vol. 15, no. 1, p. 154, 2022.

[7] W. Fang, J. Jiang, S. Lu, Y. Gong, Y. Tao, Y. Tang, and J. Liu, "A LSTM algorithm estimating pseudo measurements for aiding INS during GNSS signal outages," *Remote Sens.*, vol. 12, no. 2, p. 256, 2020.

[8] Z. Zhi, D. Liu, and L. Liu, "A performance compensation method for GPS/INS integrated navigation system based on CNN–LSTM during GPS outages," *Meas.*, vol. 188, p. 110516, 2022.

[9] S. Chen, M. Xin, F. Yang, X. Zhang, J. Liu, G. Ren, and S. Kong, "Error compensation method of GNSS/INS integrated navigation system based on AT-LSTM during GNSS outages," *IEEE Sens. J.*, 2024.

[10] G. Falco, M. Pini, and G. Marucco, "Loose and tight GNSS/INS integrations: Comparison of performance assessed in real urban scenarios," *Sensors*, vol. 17, no. 2, p. 255, 2017.

[11] G. M. Yan and J. Weng, *Strapdown Inertial Navigation Algorithm and Integrated Navigation Principle*. Xi'an, China: Northwest Univ. of Technol. Press, 2019.

[12] A. Budiyono, "Principles of GNSS, inertial, and multi-sensor integrated navigation systems," *Ind. Robot*, vol. 39, no. 3, 2012.

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, and I. Polosukhin, "Attention is all you need," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[14] A. Sherstinsky, "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network," *Physica D*, vol. 404, p. 132306, 2020.

[15] Y. Yao and X. Xu, "A RLS-SVM aided fusion methodology for INS during GPS outages," *Sensors*, vol. 17, no. 3, p. 432, 2017.

Authorized licensed use limited to: Istanbul Bilgi University. Downloaded on July 25,2026 at 23:28:21 UTC from IEEE Xplore. Restrictions apply.